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Studio Session	2

Part A: URL of the Visualisation

Paste your visualisation URL in the empty space below.

Tip: Test your URL in incognito mode to ensure it is a working URL

https://nlau0021-beep.github.io/FIT2179_nlau0021/

Part B: Domain Description

Describe your domain chosen for DV II

My domain is based on the population ageing and its consequences for public health in Malaysia. In my research, Malaysia is undergoing a quiet but rapid demographic transition: in one generation the population has nearly doubled, the median age has risen sharply, and the diseases that kill us have shifted from infectious illness to chronic, age-related conditions. This visualisation tells that story across ten linked charts, moving from a national overview down to the state level and outward to the regional ASEAN context.

Describe the Why and Who aspects of your DV II

Why. Malaysia's policy conversation about ageing is still treated as a future problem and yet 11.8% of Perak's population is already aged 65 or older, and 65% of all Malaysian deaths now come from non-communicable diseases. The visualisation reframes ageing as a present-day public-health concern with visible geographic disparities, helping readers see where the pressure already sits.

Who. The intended audience is the general Malaysian public, particularly working-age adults who may not yet realise that ageing is reshaping the health system they will rely on. No statistical training is assumed; rates and ratios are explained inline.

Describe the “What” aspect of your DV II

The visualisation combines data from two independent sources. Population data comes from the Department of Statistics Malaysia (DOSM) Population Table: States dataset, which contains annual single-year-of-age, sex, and ethnicity counts for every state from 1991 to 2025. Health data comes from the World Health Organization's Malaysia country profile, specifically the top causes of death, the share of deaths by broad cause, and the probability of premature NCD mortality. State boundaries are simplified ADM1 polygons from geoBoundaries. Several derived measures were computed from the raw DOSM file — median age by state, share aged 65 or older, working-age support ratio, and the compounded five-year growth rate — using local Python preprocessing scripts.

Describe the “How” aspect of your DV II

The story uses eight distinct idiom across ten charts, chosen to match the task at each step: stacked area for national growth, choropleth for the state-level spatial view, diverging-bar pyramids for cohort structure, small multiples for change over time, ranked bars for cause-of-death comparison, donut for proportion, multi-line for ASEAN comparison, dot strip for state ranking, and a custom demographic clock as the custom-build summary.

The demographic clock is a custom-built bubble idiom plotting each state on a two-axis grid (median age $\times$ compounded annual growth rate), with bubble size encoding total population and colour encoding the elderly share. Four labelled quadrants ("Young & growing", "Ageing & growing", etc.) turn position alone into an actionable typology — the chart's job is categorisation, not just comparison. Other custom elements include an off-scale callout for Putrajaya's outlier support ratio, a highlight-and-fade technique that picks Malaysia out from five ASEAN peers, and editorial subtitles on the small-multiples maps that surface the progression numerically as well as visually.

Part C: Sketch

A URL to a PDF document with an image of your sketch on your GitHub account.

Tip: Test your URL in incognito mode to ensure it is a working URL

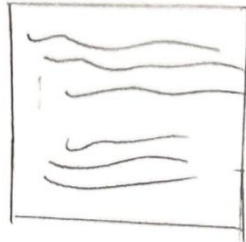
https://github.com/nlau0021-beep/FIT2179_nlau0021/blob/main/assets/FIT2179_Assignment2_Sketch.pdf

Paste a screenshot of your sketch for your DV II.

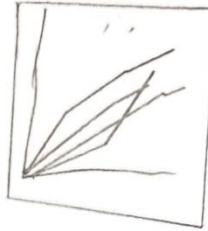
Note: Use the entire page here for the screenshot and ensure that it is clearly legible and visible

MALAYSIA: AGE & LIFE

Sub title



Population by race



sub title

Median age by state



Age state by year

1990

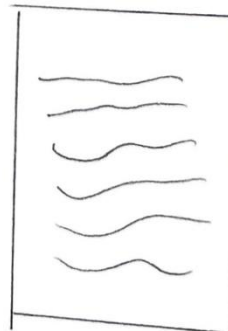
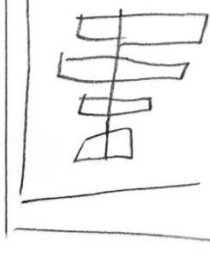
2000

2020



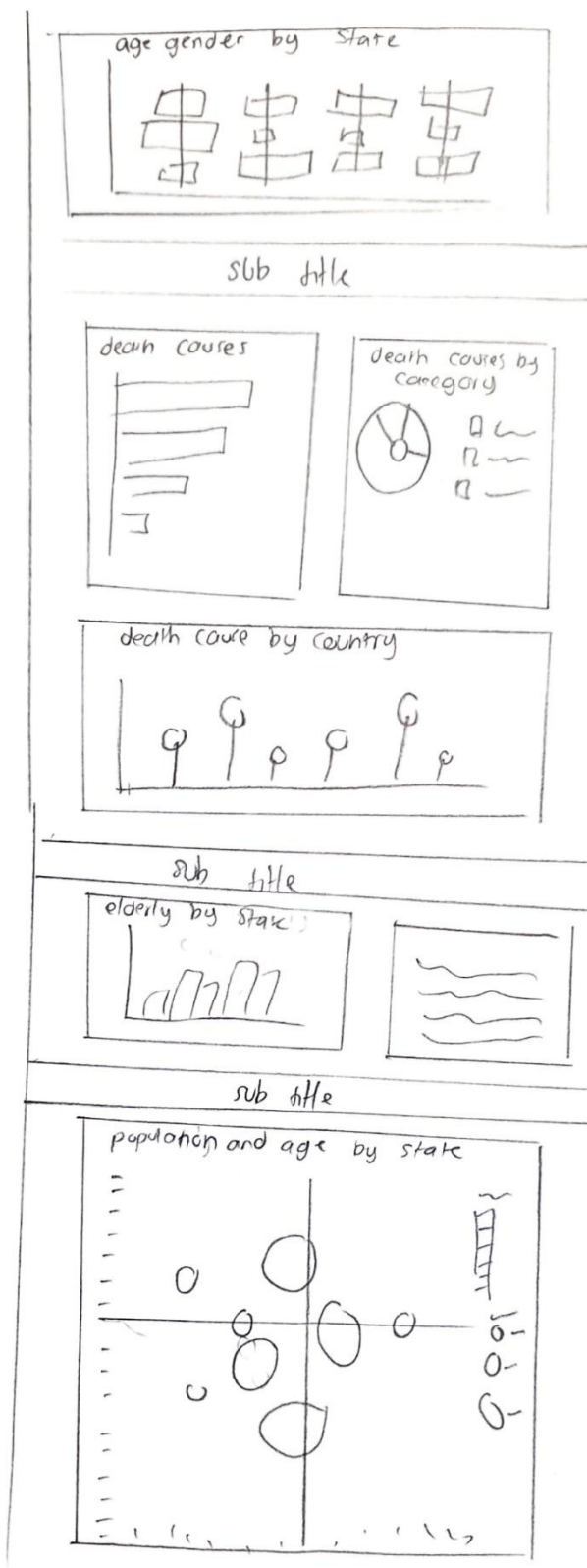
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Age by gender



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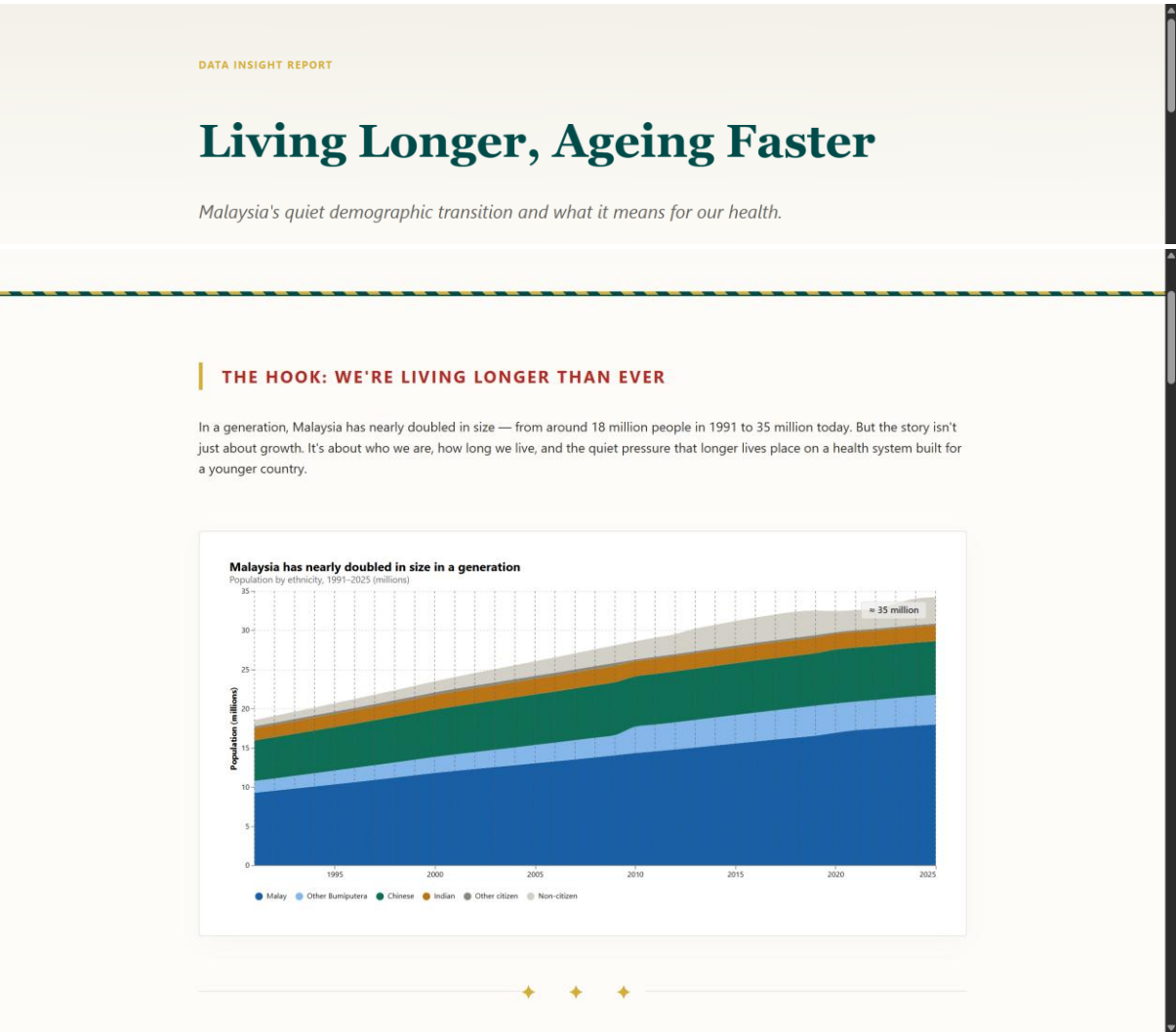


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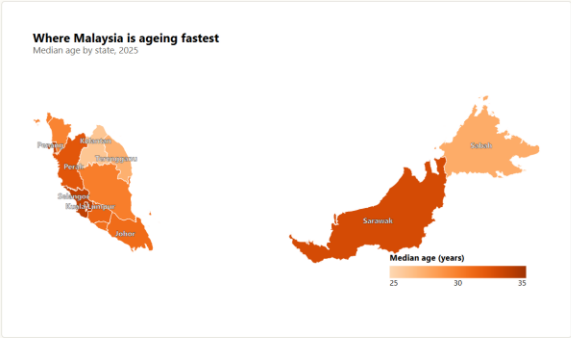
Part D: Full-size screenshot of your DV II

Paste a full-size screenshot of your DV II assignment from the URL here



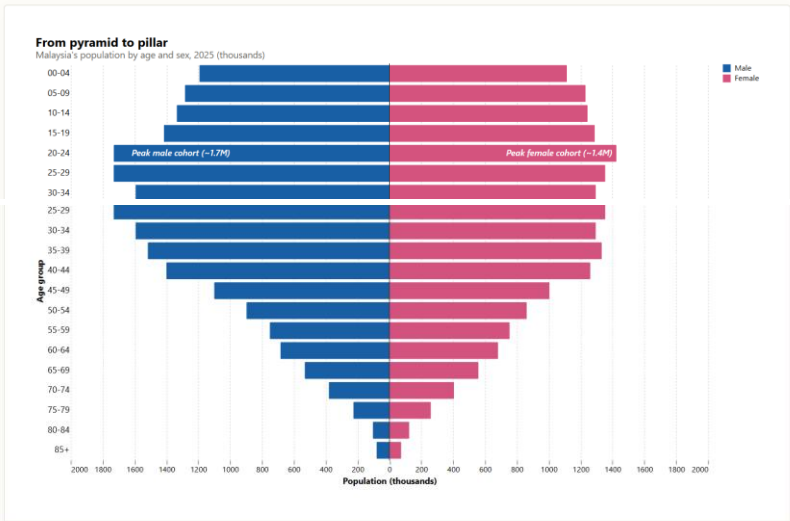
WHERE THE AGEING IS HAPPENING

The pressure of an ageing population isn't evenly spread. The peninsular west coast — Perak, Penang, Selangor, Kuala Lumpur — has aged the fastest. East Malaysia (Sabah, Sarawak) remains the youngest part of the country.



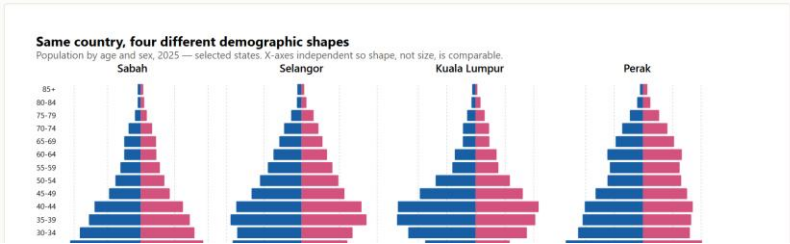
THE SHAPE OF THE POPULATION

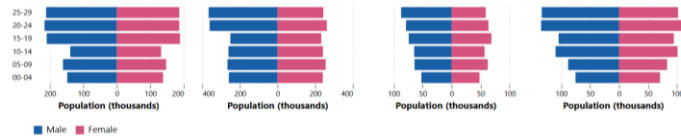
Each bar shows how many people are in a 5-year age band. Read it like a tree ring: the bands at the bottom (children) versus those above (older cohorts) tell you how the population is changing over time.



SAME COUNTRY, FOUR DIFFERENT DEMOGRAPHIC SHAPES

Zoom in on individual states and the picture becomes more varied. Sabah still looks like a textbook population pyramid — wide at the base, narrowing upward. Perak's pyramid has lost its bottom: fewer children entering each year. Kuala Lumpur bulges in the middle, dominated by working-age migrants drawn to the capital.



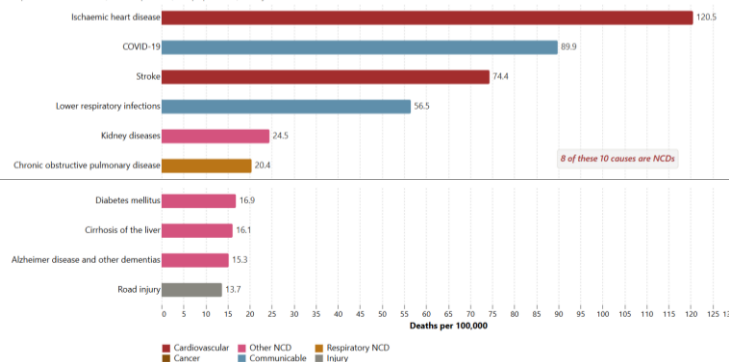


WHAT AN OLDER MALAYSIA MEANS FOR OUR HEALTH

As the population ages, the diseases that kill us shift. Today, eight of Malaysia's top ten causes of death are non-communicable — chronic illnesses that build up over decades of life.

Heart disease, COVID, stroke — what's killing Malaysians

Top 10 causes of death, deaths per 100,000 population, Malaysia 2021



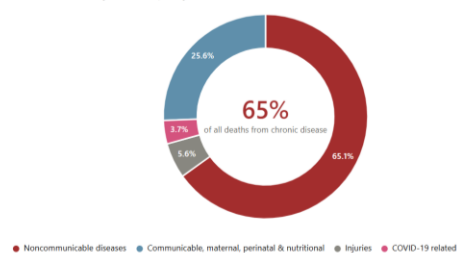
THE 65% PROBLEM

Put another way: out of every 100 Malaysians who die today, 65 die from a chronic disease. These are deaths that, in most cases, could have been prevented or delayed with earlier intervention.

That share has grown steadily over the past two decades, and is set to keep growing as the population ages.

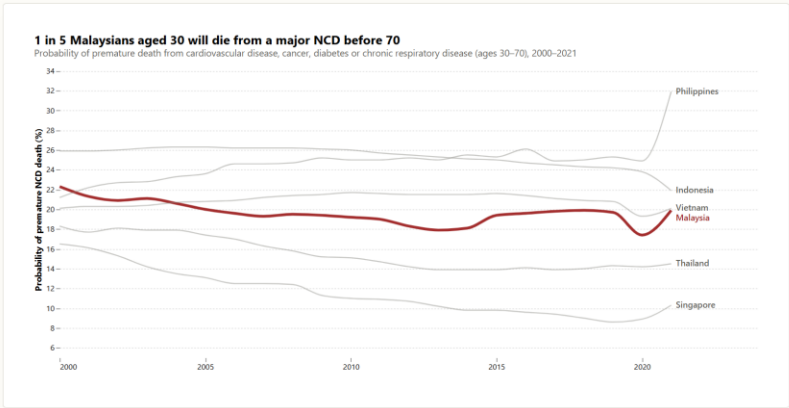
65% of deaths are now from chronic disease

Share of total deaths by broad cause, Malaysia 2021



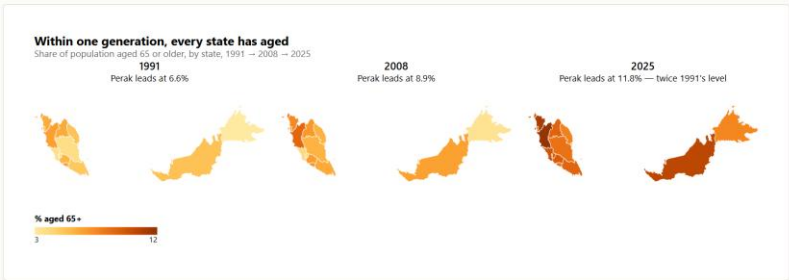
HOW DOES MALAYSIA COMPARE REGIONALLY?

The probability that a 30-year-old will die from a major chronic disease before age 70 is a single, powerful measure. Malaysia sits in the middle of the ASEAN pack — making slower progress than Singapore or Thailand, but better than Indonesia or the Philippines.



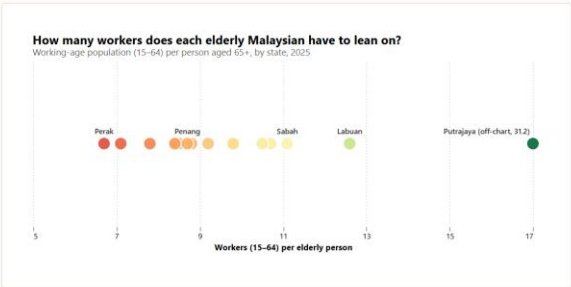
WATCH THE AGEING WAVE SPREAD

The same map, three different decades. The colour intensity shows how much of each state's population is aged 65 or older. In 1991, only Penang and Perak had a visibly older population. By 2025, every state has aged.



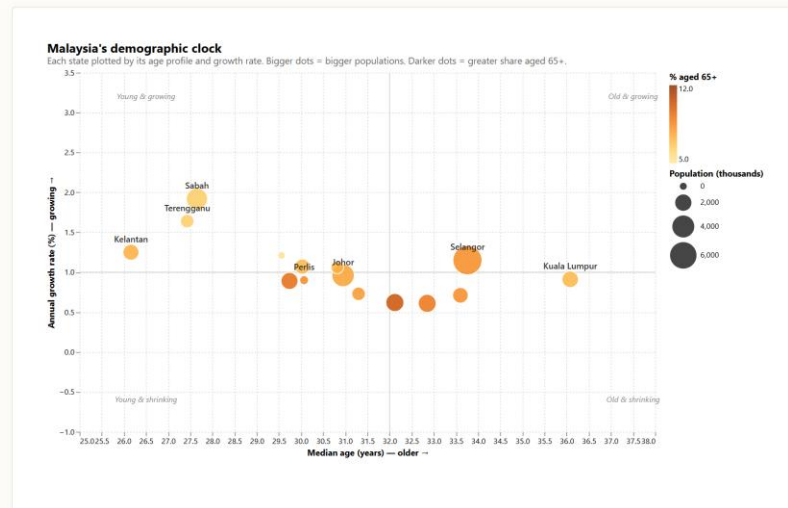
THE PRESSURE AHEAD

Behind every retiree are the working-age adults who pay the taxes, fill the jobs, and provide the care. Some Malaysian states have plenty of workers per retiree; others are already running thin.



WHERE EACH STATE STANDS ON THE DEMOGRAPHIC CLOCK

Two demographic forces shape a state's future: how old its people are today, and how fast its population is changing. Each state sits in one of four quadrants — and the quadrant tells you what challenges are coming next.



Author: Nicholas Lau | 36532819 · FIT2179 · Monash University Malaysia · May 2026

Data sources:

- Population data: [Department of Statistics Malaysia — Population Table: States](#) (CC BY 4.0)
- Health data: [World Health Organization — Malaysia country page](#) (CC BY-NC-SA 3.0 IGO)
- State boundaries: [geoBoundaries Malaysia ADM1](#) (CC BY 4.0)